

## PROFILE

Platform engineer with 10+ years building resilient document-generation pipelines, layout engines, and developer-facing template systems. Specialised in high-throughput PDF rendering, semantic authoring DSLs, and turning brittle production-ops scripts into typed, snapshot-tested libraries that scale.

## SKILLS

**Languages:** Java 21, Kotlin, Groovy, Python, SQL

**Document & Print:** PDFBox, Apache POI (DOCX/XLSX), iText, PostScript, ICC colour profiles, font metrics

**Layout engines:** Custom DSL design, semantic layout trees, pagination, snapshot testing, visual regression

**Build & infrastructure:** Maven, Gradle, GitHub Actions, JitPack, Docker, JMH benchmarking

**Testing:** JUnit 5, AssertJ, PDFBox-based PNG diff, layout-graph snapshots, mutation testing (Pitest)

**Distribution:** Maven Central, Sonatype OSSRH, GPG signing, JitPack, semantic versioning discipline

## EDUCATION

**MSc Computer Science** / University of Manchester / 2021. Distinction. Thesis: Composable layout primitives for deterministic document rendering.

**BSc Software Engineering** / Imperial College London / 2019. First-class honours. Specialisation in compilers and static analysis.

**Oracle Java Certification** / Professional track / 2023. Java 17 platform deep-dive: records, sealed types, pattern matching, virtual threads.

## ADDITIONAL

**Languages:** English (Fluent), German (Intermediate), Spanish (Basic)

**Work Eligibility:** Eligible to work in the UK and the EU

## EXPERIENCE

### Senior Platform Engineer / 2024-Present

Northwind Systems

Led the reusable document-generation platform serving billing, hiring, and reporting flows across 8 product teams. Reduced template maintenance time by 70% by retiring per-team PDF scripts in favour of one canonical engine.

### Software Engineer / 2021-2024

BrightLeaf Labs

Built backend services and production document rendering pipelines processing 2M+ documents per month. Drove the migration from iText to a custom layout engine, eliminating licensing risk and cutting p99 render latency from 1.4s to 380ms.

## SELECTED PROJECTS

**GraphCompose** - Declarative Java PDF layout engine. Semantic DSL, slot-based templates, snapshot testing. Powers production CV / invoice / proposal pipelines for hiring tools and billing systems. (Open source)

**Template Studio** - Internal tool for evaluating CV, proposal, and invoice output across 14 design presets. PNG diffing, side-by-side layout, baseline freezing.

**LayoutLint** - Static analyser that flags fragile authoring patterns (deeply nested rows, untyped offsets, implicit page breaks) before they ship to production.

**ChromeForge** - Editorial-magazine document toolkit built on GraphCompose: cinematic covers, pull quotes, multi-column flow, sidebar callouts.